

Session Facilitator Guide

"Selenium Automation Framework + GitHub Copilot Custom Agents"

Duration: 90 minutes | Audience: QE / Developer team

Session Flow at a Glance

#	Topic	Time	Cumulative
1	Target Application Walkthrough	5 min	0:05
2	Framework Architecture (README)	10 min	0:15
3	Test Classes Deep Dive	15 min	0:30
4	Configuration & Test Data	5 min	0:35
5	Page Objects & Driver	15 min	0:50
6	TestNG Suites	5 min	0:55
7	Live Test Run + Reports	10 min	1:05
8	GitHub Copilot Custom Agent	20 min	1:25
9	Q&A Buffer	5 min	1:30

PART 1 — The Target Application (5 min)

Show: <https://www.franchising.inspirebrands.com/> Talking points:

- This is a Squarespace-hosted CMS with 7 franchise brand pages
- Open the "Our Brands" dropdown → walk through all 7 brands (Arby's, Baskin-Robbins, BWW, BWW GO, Dunkin', Jimmy John's, SONIC)
- Navigate to <https://www.franchising.inspirebrands.com/arbys> — this is the page we automate
- Point out: hero section, "Why Arby's?", qualification requirements, "GET STARTED" CTA, footer
- Key challenge to mention:** Squarespace renders text inside `<div>` elements, uses curly apostrophes (U+2019), and CSS scroll animations — all of which make locators harder than they look

PART 2 — Framework Architecture via README (10 min)

Open: `README.md` Walk through each section: Architecture diagram — read the layered diagram top-down:

``

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TestNG Suite XML → Test Layer → Page Object Layer → Factory → Infrastructure → Config

` Project structure tree - explain each folder:

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- `src/main/java/com/inspire/` - framework code (not tests)

- `src/test/java/com/inspire/tests/` - test classes

- `src/test/resources/` - config + brand properties

- `testng-suites/` - suite XML files

- `mvnw.cmd` - Maven Wrapper (no system Maven needed)

Design patterns table - briefly name each pattern (will demo in code shortly) **Adding a New Brand - 4 steps** - emphasise: zero existing files modified

PART 3 – Test Classes Deep Dive (15 min)

3a. AbstractBrandTest.java (5 min)

Open: `src/test/java/com/inspire/tests/brands/AbstractBrandTest.java` **Talking points:**

- This is an **abstract class** - never runs directly
- Contains TC-B-01 through TC-B-12 - the 12 tests every brand gets for free

- `getBrandPage()` is the **Template Method Pattern** hook - subclasses provide the page object

- `@BeforeMethod openBrandPage()` - calls `brandPage.open()` before each test

- Point to `brandPage.getBrandDisplayName()` - never hardcoded brand names

- **Audience question:** "If I add Dunkin' tomorrow, how many test lines do I write to get these 12 tests running?" → **Answer: zero**

3b. ArbysTest.java (5 min)

Open: `src/test/java/com/inspire/tests/brands/ArbysTest.java` **Talking points:**

- extends `AbstractBrandTest` → inherits all 12 common tests automatically

- `getBrandPage()` `returns` `new ArbysPage(driver)` - one line to wire the brand

- `arbysPage()` `private helper` - avoids repeated casts

- TC-A-01 through TC-A-09 - Arby's-specific validations only

- Point to TC-A-05/TC-A-06 - \$500,000 `liquid assets`, \$1,000,000 `net worth`

- **Total test coverage:** 12 common + 9 Arby's-specific = 21 tests from 1 class

3c. `HomePageTest.java` (5 min)

Open: `src/test/java/com/inspire/tests/HomePageTest.java` **Talking points:**

- Tests the home page / independently - brand-agnostic

- TC-H-01: URL check, TC-H-02/03: hero heading, TC-H-05: GET STARTED navigation

- Point to `homePage.isBrandInDropdown(Brand.ARBYS.getDisplayName())` - TC-H-07

- Every test calls `logStep()` and `logPass()` - these auto-capture screenshots (show a screenshot in the report later)

PART 4 - Configuration & Test Data (5 min)

4a. `config.properties`

Open: `src/test/resources/config.properties` **Talking points:**

- Controls browser, headless mode, base URL, active brand, timeouts

- Show system property override priority: `-Dheadless=true` beats the file

- Demo commands:

`powershell`

`.\mvnw.cmd clean test -P arbys -Dbrowser=firefox -Dheadless=true`

4b. brands/arbys.properties

Open: `src/test/resources/brands/arbys.properties` **Talking points:**

- **Brand-specific data overlay** – loaded on top of `config.properties` **when** `active.brand=arbys`
- **Shows** `brand.liquid.assets=500,000` , `brand.net.worth=1,000,000`
- Show the pattern: every brand gets its own properties file
- New brand = new properties file, zero framework changes

PART 5 – Page Objects & Driver (15 min)

5a. ArbysPage.java (4 min)

Open: `src/main/java/com/inspire/pages/brands/ArbysPage.java` **Talking points:**

- extends `AbstractBrandPage` – **Liskov Substitution in action**
- `@FindBy` **annotations** – **Page Factory pattern**, no `driver.findElement()` **in test code**
- `getBrandPagePath()` **returns** `"/arbys"` – **only brand-specific thing**
- **Point out apostrophe handling in XPath:** `contains(., 'Why')` and `contains(., 'Arby')` – **not** `"Why Arby's"`
- `isLiquidAssetsRequirementDisplayed()` → **ties back to TC-A-05**

5b. AbstractBrandPage.java (4 min)

Open: `src/main/java/com/inspire/pages/brands/AbstractBrandPage.java` **Talking points:**

- **Common** @FindBy **locators shared by ALL brands** – helpSectionHeading , whatNextHeading , stepApply through stepSign

- implements IBrandPage – **Dependency Inversion**, tests depend on the interface

- clickGetStartedAndGetUrl() – **show the** waitForUrlContains() + href fallback (explain P-03 pitfall)

- isGetStartedButtonDisplayed() – **scrolls before checking visibility** (explain P-02 pitfall)

5c. **HomePage.java** (3 min)

Open: src/main/java/com/inspire/pages/HomePage.java **Talking points:**

- implements INavigable – **Interface Segregation**: navigation methods separate from brand-page methods

- hoverOver(ourBrandsMenuTrigger) – **JavaScript hover for Squarespace dropdown**

- brandsSectionHeading **locator** – **show the split** contains() for "Inspire's"

5d. **DriverManager.java** (4 min)

Open: src/main/java/com/inspire/driver/DriverManager.java **Talking points:**

- ThreadLocal – **one WebDriver per thread for safe parallel execution**

- initDriver() / getDriver() / quitDriver() – **clear lifecycle**

- **Chrome options:** --headless=new , --no-sandbox , --window-size=1920,1080

- WebDriverManager.chromedriver().setup() – **no manual ChromeDriver download needed**

- **Called from** BaseTest.setUpMethod() / tearDownMethod() – **developers never call this directly**

5e. WaitUtils.java (bonus 3 min)

Open: src/main/java/com/inspire/utis/WaitUtils.java Talking points:

- waitForVisibility() , waitForClickability() , scrollIntoView() - explicit waits only, implicit wait = 0
- scrollIntoView() is the key fix for Squarespace scroll animations
- waitForUrlContains() - used in clickGetStartedAndGetUrl() fallback

PART 6 – TestNG Suite XML Files (5 min)

Open both: testng-suites/testng-arbys.xml and testng-suites/testng-all.xml Talking points for testng-arbys.xml :

- Two nodes: Smoke Tests + Full Regression - run in order
- - selective execution
- Maven profile -P arbys points Surefire at this file

Talking points for testng-all.xml :

- Shows the extensible structure: add a block for each new brand
- Future brand stubs are commented in - one uncomment to enable

Show the Maven profiles in pom.xml :

- -P arbys → runs testng-arbys.xml
- -P all-brands → runs testng-all.xml
- -Dgroups=smoke → runs only smoke group across whichever suite

PART 7 – Live Test Run + Report Demo (10 min)

```
Run in terminal: powershell
.\mvnw.cmd clean test -P arby's -Dgroups=smoke

While tests run, narrate:
• Maven downloads dependencies on first run (show the download progress)
• WebDriverManager silently downloads ChromeDriver
• Chrome browser opens, cookie banner dismissed automatically
• Tests execute – point out the browser navigating the Arby's page
```

```
After run completes, open the report:
test-output\extent-reports\report.html
```

```
Walk through the report:
• Dashboard: pass/fail count, timeline bar
• Click TC-B-03 → show step-by-step log with inline screenshots
• Click a failed test (if any) → show "Screenshot at failure" screenshot
- "This tells you the exact browser state when it failed"
• System/Environment panel: Browser, Active Brand, Java version, OS
```

PART 8 – GitHub Copilot Custom Agent (20 min)

8a. Tools & Their Permissions (5 min)

```
Open: .github/agents/Inspire-Franchise-Selenium-Automation-Framework-Agent.agent.md Show the YAML frontmatter: yaml
tools: [vscode/getProjectSetupInfo, execute/runTests, execute/testFailure,
read/readFile, edit/createFile, search/codebase, web/fetch, ...]

Group them for the audience:
```

Category	Tools	What they enable
vscode/*	installExtension, memory, runCommand	IDE integration, remembers session state
execute/*	runInTerminal, runTests, testFailure	Run Maven, read test failures
read/*	readFile, problems	Read any Java/XML/properties file
edit/*	createFile, editFiles	Create DunkinPage.java, update factory
search/*	codebase, textSearch, usages	Navigate the entire codebase
web/*	fetch, openBrowserPage	Fetch live site DOM for new locators

Key point: "The agent can open the actual franchising website, read the DOM, and write the correct @FindBy locator for a new brand – without you having to inspect the page yourself."

8b. MANDATORY Rules – Why They Matter (5 min)

```
Still in the agent file, show the three MANDATORY sections: MANDATORY – Always Use clean test :
```

"Every time a developer types mvnw test without clean and hits Cannot instantiate class , they lose 20 minutes. This rule costs zero tokens and saves hours."

```
MANDATORY – Cookie Banner:
```

"Maven Surefire starts a fresh Chrome with no cookies. The banner appears every single time. Without `dismissCookieBannerIfPresent()` , every test fails at the click step."

MANDATORY – XPath Rules:

- `Open` `ArbysPage.java` , `show` `whyArbysHeading` `locator`

"The apostrophe in 'Arby's' is a Unicode curly quote (U+2019). Ask Copilot without this rule – it writes "Why Arby's" – instant `SyntaxError` . With the rule – it always splits into two `contains()` calls."

8c. Project Structure in the Agent (3 min)

Show the Project Structure section in the agent file:

"The agent knows every file's responsibility. When you say 'add a Sonic brand', it creates `SonicPage.java` , adds case SONIC: to the factory, creates `SonicTest.java` , and creates `testng-sonic.xml` – in the right packages, with the right class signatures."

"Without this section, Copilot guesses the package structure. With it, the first answer is correct."

8d. Common Pitfall Demo – GET STARTED Button (7 min)

The most impactful demo – live with the agent selected in Copilot chat. Switch to Copilot chat, select the agent, then ask:

"The GET STARTED button on the Arby's page is not clicking correctly. What could be wrong?"

Agent's answer should reference P-03:

- Multiple hidden nav copies of the button exist

- `Locator` needs `not(ancestor::header)` , `not(ancestor::nav)` , `contains(@href,'franchise-with-us')`

- `clickGetStartedAndGetUrl()` already has the `waitForUrlContains()` + `href` fallback

Then ask:

"Add Baskin Robbins brand to this framework"

Watch the agent:

`Create` `BaskinRobbinsPage.java` in the correct package

`Add` case BASKIN_ROBBINS: to `BrandPageFactory.java`

`Create` `BaskinRobbinsTest.java` extending `AbstractBrandTest`

`Offer to create` `testng-baskin-robbins.xml`

Close with:

"The 12 common tests from AbstractBrandTest run automatically for the new brand. The agent saved roughly 4 hours of boilerplate and zero existing files were changed – that's the Open/Closed Principle enforced by the agent, not just documented."

Q&A Prompts (to fill time if needed)

- "What would break if we used `mvn test` instead of `.\mvnw.cmd clean test`?"
- "Why is there a `ThreadLocal` in `DriverManager`? What problem does it solve?"
- "The `@FindBy` annotation uses `normalize-space(.)` – what's the dot?"
- "How do we add a CI pipeline trigger for just the smoke group?"
- "What's the difference between Ask, Plan and Agent mode in Copilot?"

Pre-Session Checklist

- ☐ Browser open at <https://www.franchising.inspirebrands.com/>
- ☐ VS Code open at `C:\code\Inspire\InspireFranchises`
- ☐ Terminal ready in project root
- ☐ Copilot agent pre-selected: Inspire-Franchise-Selenium-Automation-Framework-Agent
- ☐ Test run done once beforehand (Maven dependencies already downloaded – live run will be fast)
- ☐ Previous Extent report open as reference in browser
- ☐ Presentation slides open: `team-presentation\GitHub-Copilot-for-Automation-Frameworks.pptx`